

Addendum to “From the Warp/Wormhole Interface to a Throat-Supported Shift Rail”: Corrections, Geometry Clarification and Current State

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This addendum corrects the May 2026 paper [1]. Work since its release shows three things: the paper’s reduced model describes a two-ended traversable wormhole; several of its readings belong to a rail through one space; and its service sequence, applied to its throat, demands a stress that no source with a rest frame supplies. The two service rules the paper found carry forward into a one-space design that passes that admissibility test. The technical disclosure cited here is the version at a fixed commit [2]; a later revision will follow the source evaluation.

1. The geometry of the paper

The reduced diagnostic model uses the spherically symmetric form

$$ds^2 = -\alpha^2 ds^2 + \gamma_u (dl + \beta^l ds)^2 + \gamma_{\Omega\Omega} d\Omega^2,$$

with the rail coordinate l as its radial direction. Its implementation sets the areal radius by the Ellis profile $R^2 \propto l^2 + R_{th}^2$, a throat of radius $R_{th} = 1.75$ at $l = 0$. A later revision held $R = 1.75$ along the whole track, with flared ends. The geometry is therefore a two-ended, traversable wormhole of Morris–Thorne type. Its lapse stays positive and finite, so light and slow travelers cross the throat directly. The metric leaves open whether the two flat regions belong to one universe. The paper’s system is thus an active rail laid through this throat, and its shortcut combines the throat’s topology with the shift along it.

The shift also fixes the construction’s place in the wormhole–warp correspondence [4]. There, a localized warp bubble crossing a horizon-free throat meets a curvature singularity. The rail’s shift is instead the throat’s own spherically symmetric radial field, written in the proper-distance coordinate with a non-unit lapse. That places it in the regular spherical class of the correspondence, a class the paper left unnamed.

2. Issues in the paper

1. **Unstated topology.** The paper presents the throat as infrastructure that the service prepares and relaxes. It also draws on the prepared-route picture of the Krasnikov tube, a route through ordinary space. Its geometry, however, is a wormhole with two asymptotic ends, and the paper leaves that topology and its consequences unstated.
2. **Incomplete source premise.** Supplied sources can provide the stress that holds a throat open. Forming a two-ended space from ordinary space requires a change of spatial topology, which in a compact region brings closed timelike curves [6]. The paper’s conditional feasibility claim therefore rests on a geometry that sources alone cannot produce.
3. **Inadmissible stress.** The paper’s screen tests packet timelikeness, a stationary monitor, radial light speeds and bundle compression. The algebraic type of the demanded stress lies outside it. The Hawking–Ellis classification applied later [5, 3] shows that the paper’s own relaxation step demands a Type IV layer. That step is the support decompression, acting on the throat’s varying areal radius. It splits the two radial null energies into opposite signs, which leaves the stress with no timelike eigenvector, so no source with a rest frame can supply it. Decompression alone produces 874 Type IV points in a controlled run, and the full operating geometry of the same service family produces 13,587 on its certification map.
4. **Mismatched packet readings.** In the service family behind the paper, the packet windows advanced at unit coordinate speed while the packet-norm test used the carry speed U/B . The packet screen therefore combined two different readings of the packet’s motion [3].
5. **Unseparated transport.** The paper describes traversal as a timed rail service and leaves the throat’s shortcut and the shift’s transport unseparated. The disclosure [2] then rated the service against an exterior null signal between endpoints taken a rail-coordinate distance apart. Those endpoints lie in separate asymptotic regions. Its service-time ratios, 2.569 and 1.233, therefore compare throat traversal with a light path the geometry lacks, and the mismatch of item 4 enters both. They are proxies, and the geometry defines no arrival lead.
6. **Overstated confidence.** The paper concludes that its reduced results “increase confidence” in a conditional engineering possibility. Given items 1–3, that confidence applies to its two service rules alone. The geometry on which they were tested requires a change of topology and demands inadmissible stress.

3. What carries forward

The paper’s two service rules hold. Transport shift stays inside its support envelope, and the packet is caught before release. Its packet-centred division of labour also holds: the packet worldtube is the passenger object, and infrastructure bears the gradients. The service sequence, prepare $\rightarrow$ carry $\rightarrow$ catch $\rightarrow$ release $\rightarrow$ relax $\rightarrow$ reset, now runs as a literal schedule.

4. Current state

The current design realizes the rail as a track in one flat space, with no throat and unit stretch [3]:

$$ds^2 = -\alpha^2 d\sigma^2 + (dz + \beta d\sigma)^2 + dr^2 + r^2 d\phi^2, \quad \rho = -\frac{1}{32\pi} \left(\frac{\partial_r \beta}{\alpha} \right)^2.$$

The Hamiltonian constraint on the flat slices fixes this energy density. The service region $r \leq 1.75$ is the packet-safe region. There the shift carries the packet along one prescribed path. It enters at $0.9c$, crosses a $2.1c$ lane, slows to $0.95c$ across the lane’s edge and is released. The shift sits inside a convex lapse plateau of e^4 , whose windows open with the lapse leading and lagging the shift. A transverse boundary layer returns the fields to Minkowski values by $r = 13.25$. Its lapse sheath of e^1 travels with the packet on a schedule, so flat space remains behind each transit, and sources placed along the lane switch on as the packet arrives.

- **Arrival.** The packet reaches the track end at $\sigma = 3.34$. Light through flat space from the same entry event arrives at $\sigma = 5.0$. That gives the packet a mean speed of $1.35c$, set entirely by the shift and lapse.
- **Packet.** Its worldtube is timelike from approach to arrival. In the lane the packet moves at $0.016c$ or less relative to local normal observers, and its clock runs at most 55 times exterior time.
- **Admissibility.** Over $\sigma \in [-8, 10]$ and $|z| \leq 10$, every non-vacuum boundary-layer point is Hawking–Ellis Type I, including under refinement. No resolved crossing carries a Type IV band, and the exterior is exactly vacuum.
- **Demand.** During a transit the demand is a tension without energy on the shoulders of the lapse profile, local to the packet: peak stress 2.9 and peak negative energy 1.9×10^{-4} , in units where the service region has radius 1.75. At one unit per metre, the peak null-energy deficit integrates to about $0.1 M_\odot$ and the peak stress is 3.5×10^{44} Pa. Both are independent of route length.

The next stage evaluates candidate source families against this demand.

References

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